PLC Memory Organization

Allen Bradley PLC Processors

1 Program Interchangeability

Non-Interchangeability of PLC Programs

PLC programs are **not interchangeable** among different models or manufacturers. This is due to several fundamental engineering differences:

- **Memory Organization:** Differences in how the CPU allocates and manages storage.
- **Memory Layouts:** Unique physical and logical mapping of registers and bits.
- **File Structures:** Varying ways of organizing logic and data blocks.
- **Addressing Methods:** Formats for referencing I/O and internal memory vary significantly.
- **Instruction Sets:** Variations in available commands, syntax, and functional behavior.

Syntax Variation Example

Comparing a simple Normally Open (NO) contact and assignment:

- **Allen Bradley (SLC 500):** Uses 'XIC' (Examine If Closed) and addressing like 'I:1/0'.
- **Siemens (S7-1200):** Uses standard IEC symbols and addressing like 'I0.0'.

2 Allen Bradley Memory Sections

The processor memory is logically divided into **two primary sections**:

1. **Program Files:** Contains the executable control logic and system routines. These define the machine's operations and account for the majority of total memory.
2. **Data Files:** Acts as the "working memory," storing real-time information, status flags, and numerical values used by the program.

File Number	Type	Description
File 0	System Functions	Holds processor type, passwords, diagnostics, and I/O config.
File 1	Reserved	Used by the processor internally; *not accessible to programmer.*
File 2	Main Ladder	The primary user logic, scanned repeatedly during operation.
File 3–255	Subroutines	Modular subprograms called from File 2 for reusable logic.

Table 1: Program File Organization

3 Program Files Breakdown

Subroutine Call Example

To keep the main program clean, complex tasks are moved to subroutines.

- **Instruction:** Use the `JSR` (Jump to Subroutine) command in File 2.
- **Target:** Reference a file number between 3 and 255 (e.g., `JSR U:3`).

4 Data Files and Storage

Data files act as the "working memory," storing real-time information that the logic program reads from and writes to.

Data Storage Architecture

- **Structured Memory:** The PLC stores all data in structured memory areas called **Data Tables** or **Data Files**.
- **16-Bit Operations:** Memory organization is primarily based on **16-bit** operations (words).
- **Addressing:** Data is accessed by specifying a unique **memory address** which points to a specific file, word, or bit.

File Number	Type	Description
File 0	Output (O)	ON/OFF state of physical output terminals.
File 1	Input (I)	ON/OFF state of physical input terminals.
File 2	Status (S)	System flags, scan time, error codes, and diagnostics.
File 3	Bit (B)	Internal relay logic; holds intermediate logic states.
File 4	Timer (T)	Preset, accumulated values, and status bits (EN, DN, TT).
File 5	Counter (C)	Preset, accumulated counts, and status bits (CU, CD, DN).
File 6	Control (R)	Tracks position/length for shift registers and sequencers.
File 7	Integer (N)	16-bit signed integers ($-32{,}768$ to $32{,}767$).
File 8	Float (F)	32-bit floating-point numbers (decimal values).
File 9–255	User-Defined	Custom data types, arrays, and application storage.

Table 2: System-Defined Data Files

5 I/O Image Tables

The PLC uses dedicated memory areas to mirror the state of physical world devices.

5.1 Input Image Table File

This is a software mirror of the physical input status stored in memory.

- **Operation:** Each connected input has a corresponding bit in the table.
- **I/O Scan:** During the scan cycle, the CPU reads all input terminals.
- **Logic:** If voltage is detected, a **1 (ON)** is stored; if no voltage, a **0 (OFF)** is stored.
- **Real-time:** The table is updated continuously to reflect the actual state of field devices.

5.2 Output Image Table File

This table holds the desired ON/OFF states that the PLC logic intends to send to the physical world.

- **Operation:** Each connected output has a corresponding bit in this table.
- **Ladder Logic:** The program determines states and writes **1** or **0** to the table bits.
- **Update:** During the I/O update phase of the scan, physical output terminals are set to match the table.

6 I/O Addressing Format

To identify specific I/O points in memory, Allen Bradley uses a standardized **3-part format**:

```
[Type] :  [Slot] / [Terminal]
```

- **Part 1 (Type):** `I` for input or `O` for output. A **colon (:)** separates the module type from the slot number.
- **Part 2 (Slot):** Specifies the physical or logical slot where the I/O module is installed. A **forward slash (/)** separates the slot from the terminal.
- **Part 3 (Terminal):** The specific terminal screw on the module where the external device is wired.

Addressing Breakdown

- **I:1/0** → **Input** module at Slot **1**, Terminal **0**.
- **O:2/5** → **Output** module at Slot **2**, Terminal **5**.
- **B3:0/1** → Binary/Bit file **3**, Word **0**, Bit **1**.
- **T4:0.ACC** → Timer file **4**, Timer **0**, Accumulated value.

Memory organization is the foundation of efficient PLC programming.